

Lab Assignment – 1

Simulation of Router Queueing Delay

Objective

Develop a C program to simulate a simple packet-switched network consisting of a source, a router with a bounded FIFO queue, and a destination. Analyze the impact of traffic intensity on queueing delay, end-to-end delay, and packet loss.

Network Model

Source —————▶ **Router (Bounded FIFO Queue)** —————▶ **Destination**

The source generates packets according to a Poisson process (exponentially distributed inter-arrival times). The router processes incoming packets, stores them in a finite FIFO queue, and forwards them over a fixed-bandwidth output link. Packets arriving when the queue is full are dropped.

Input Parameters

Your program should accept the following input parameters:

1. Packet length (bytes)
2. Source-to-router link bandwidth (bps)
3. Router-to-destination link bandwidth (bps)
4. Source-to-router propagation delay (seconds)
5. Router-to-destination propagation delay (seconds)
6. Router processing delay (seconds per packet)
7. Router queue capacity (number of packets)
8. Packet generation rate, λ (packets/second)
9. Number of packets to simulate
10. Random number seed

Simulation Requirements

For every accepted packet, compute the following:

- Queueing delay

- Transmission delay
- Propagation delay
- Processing delay
- Total end-to-end delay

Packets arriving when the router queue is full must be discarded.

Traffic Intensity

The traffic intensity of the output link is given by

$$\rho = (\lambda * L) / R$$

where

- λ = Packet generation rate (packets/second)
- L = Packet length (bits)
- R = Router-to-destination link bandwidth (bits/second)

Conduct experiments for the following traffic intensities by varying only the packet generation rate:

$\rho = 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0, 1.1, \text{ and } 1.2$

Output

Generate a CSV file containing the following information for each experiment:

- Traffic intensity (ρ)
- Packet generation rate (λ)
- Number of packets generated
- Number of packets delivered
- Number of packets dropped
- Packet-drop probability
- Average queueing delay
- Average end-to-end delay
- Maximum queue occupancy

Graphs

Plot the following graphs using a spreadsheet:

1. Traffic Intensity (ρ) vs. Average Queueing Delay
2. Traffic Intensity (ρ) vs. Average End-to-End Delay
3. Traffic Intensity (ρ) vs. Packet-Drop Probability

Analysis Questions

Based on your simulation results, answer the following questions:

1. Which delay components remain approximately constant as traffic intensity increases?
2. Why does the queueing delay increase rapidly as the traffic intensity approaches $\rho = 1$?
3. What happens when $\rho > 1$? Explain its effect on packet delay and packet loss.
4. How does router queue capacity influence:
 - Average packet delay?
 - Packet-drop probability?

Submission details:

Show the C source code, the generated CSV file, and the spreadsheet containing the plots.

Prepare a brief report discussing your observations, which may be submitted later.